

AI

COMPUTER VISION BOOTCAMP · FINAL PROJECT

GuessWhat

Draw it. Guess it **before the AI does.**

Razan Almasoud

Joud Alaskar

Hayat Fageeh

HUMAN

Human vs AI: who guesses first?

HUMAN

A contestant watches the drawing and races to say the word.

VS

AI

Our CNN looks at the same drawing and guesses live, with a confidence score.

ONE PLAYER DRAWS

in the air with a hand on the webcam, or **with a pen** on white paper.

03 · HOW THE GAME WORKS

5 rounds, 60 seconds each

1

Pick a mode

Camera: draw in the air.
Pen: draw on a white page.

2

Draw

One player draws the secret word before the 60-second timer ends.

3

Race

The AI shows live guesses with confidence bars. Letter hints help the human.

4

Score

After 5 rounds, scores go to the leaderboard.

CAMERA MODE GESTURES

Pinch

Draw (green dot)

Fingers apart

Pen up, move freely (red dot)

Open palm, held

Clear the drawing

Train once, play live

Train • Google Colab, Python



Play • in the browser, JavaScript



No server: the video never leaves the player's device.

05 · ROUND 1

Let's play!

HUMAN

vs

AI

One drawing · 60 seconds · who guesses first?

▶ [Play now · guesswhat-five.vercel.app](https://guesswhat-five.vercel.app)

100,000 doodles from Quick, Draw!

20

classes (v1 had 5)

5,000

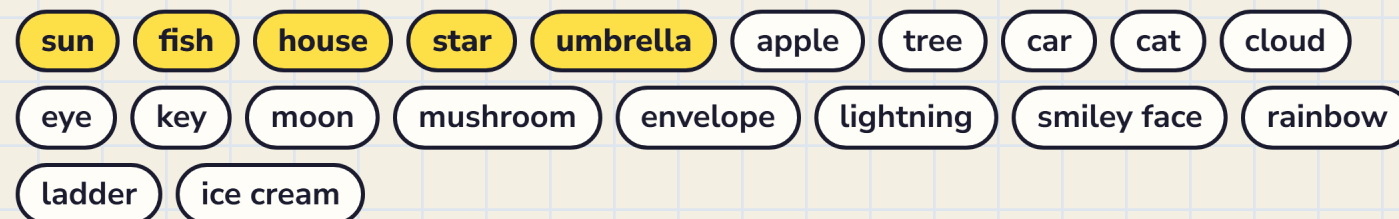
drawings per class, 28 ×
28 grayscale

80%

80,000 train · 10,000 val
· 10,000 test

Why these? Easy to draw in the air in a few strokes, and different from each other. No flower or snowflake: they look like sun and star.

Yellow = the 5 classes from v1 · white = 15 new



Smaller image, deeper meaning

Layer	Output	What it does
Input + augmentation	$28 \times 28 \times 1$	Training only: tilt $\pm 29^\circ$, shift ~ 3 px, zoom 10%
Conv 32 + MaxPool	$14 \times 14 \times 32$	Simple strokes: edges, curves
Conv 64 + MaxPool	$7 \times 7 \times 64$	Parts: corners, circles
Conv 128 + MaxPool	$3 \times 3 \times 128$	Shapes: rays, roof, tail
Flatten + Dropout 0.4	1,152	Switch off 40% of neurons while training
Dense 128	128	Combine all features
Dense 20 + softmax	20	One probability per class, sums to 100%

242,836

parameters · 948.58 KB

Small on purpose

Trains in seconds and runs in real time inside the game.

Adam · sparse categorical cross-entropy

EarlyStopping picked epoch 14

94.35%

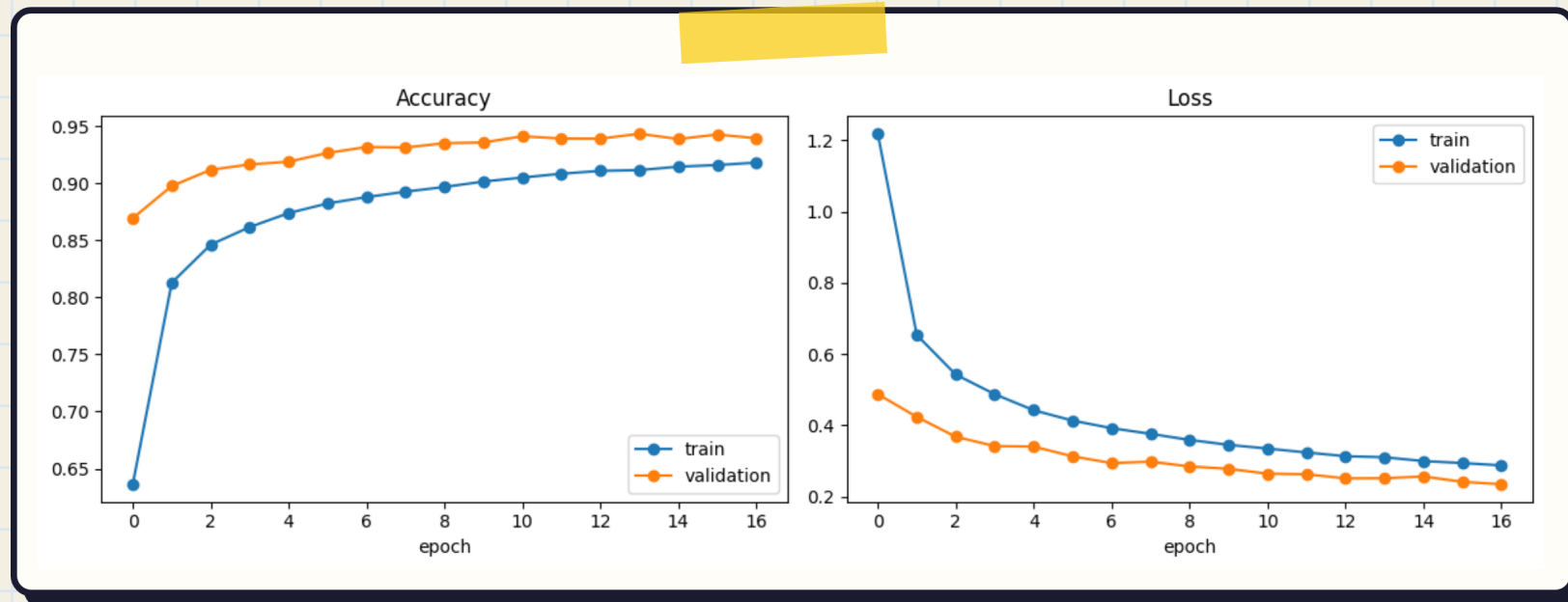
best validation accuracy,
epoch 14

17

epochs run out of a max of 20

625

updates per epoch
(batch 128)



EarlyStopping watches val_accuracy: 3 epochs with no new best → stop, and go back to the best weights.

*Validation above training? Training sees augmented drawings with dropout on.
Not overfitting!*

09 · RESULTS

The honest score: the test set

93.96%

test accuracy on 10,000 drawings never seen in training · 604 wrong

Easiest (F1)

rainbow 0.99
envelope 0.98

Hardest (F1)

cat, key, moon 0.90
mushroom recall 0.88

v1: 97.0% on 5 classes → v2: 93.96% on 4× more classes

Confusion matrix (test set)

True label \ Predicted label	sun	fish	house	star	umbrella	apple	tree	car	cat	cloud	eye	key	moon	mushroom	envelope	lightning	smiley face	rainbow	ladder	ice cream
sun	471	1	0	6	1	2	3	1	5	0	3	0	4	1	0	2	0	0	0	0
fish	0	463	0	1	0	1	0	9	9	3	1	8	2	1	1	0	0	0	1	0
house	0	1	487	2	0	1	1	4	0	0	1	0	0	1	0	0	0	0	0	1
star	0	3	2	475	2	0	1	1	5	0	1	0	3	0	1	4	1	0	1	0
umbrella	1	1	0	1	483	0	3	0	1	0	0	0	2	5	0	2	0	0	0	1
apple	0	2	0	1	0	477	0	0	5	0	2	4	7	0	0	0	1	0	0	1
tree	0	0	4	3	3	1	459	2	1	2	0	2	2	15	0	1	0	0	0	5
car	0	3	2	1	0	1	0	481	6	1	0	3	1	1	0	0	0	0	0	0
cat	1	7	0	9	0	0	2	3	461	1	1	6	3	1	0	0	2	0	0	3
cloud	0	8	1	1	1	4	1	5	2	464	1	2	7	0	0	1	1	0	0	1
eye	4	5	1	2	1	1	1	5	4	1	466	0	8	0	0	0	1	0	0	0
key	0	7	1	3	1	1	3	5	3	2	1	450	6	2	1	5	2	0	1	6
moon	1	2	1	1	0	0	0	0	3	2	6	4	472	0	0	1	7	0	0	0
mushroom	0	0	0	2	18	0	17	2	3	0	0	7	3	442	1	3	1	0	0	1
envelope	0	1	3	2	0	1	0	0	1	1	1	0	0	0	487	1	0	0	1	1
lightning	0	2	1	8	0	1	5	5	6	3	0	5	4	2	1	450	0	2	2	3
smiley face	0	0	1	1	0	1	0	3	2	0	7	0	21	0	0	1	462	1	0	0
rainbow	0	0	1	0	0	0	0	2	0	0	0	0	0	0	0	3	0	493	1	0
ladder	0	0	5	3	0	0	0	0	5	0	0	3	1	0	1	9	1	1	469	2
ice cream	0	1	1	0	1	0	5	0	0	0	0	4	1	1	0	1	0	0	1	484

Does the CNN earn its place?

Random guess



5%

Logistic regression



70.4%

Our CNN



93.96
%

Test accuracy on the same 10,000 test drawings · +23.56 points from convolutions

WHY LOGISTIC REGRESSION?

Simplest fair model

The standard first model for classification.
One line of scikit-learn.

Same data, same split

Trained on the same 80,000 drawings,
scored on the same test set.

No convolutions

Sees 784 separate pixels, not shapes. The
gap = what convolutions add.

11 · WHAT THE MODEL GETS WRONG

Is a smiley face a moon?



Most confused

True → guess	Times
smiley face → moon	21
mushroom → umbrella	18
mushroom → tree	17
tree → mushroom	15

Label noise

The label is the word the player was **asked** to draw, not what they drew: written words, stick figures, scribbles.
Fix idea: keep only **recognized: true** drawings.

A threshold for "I'm not sure"

The model always picks one of the 20 classes, even for a scribble. If its top probability is below the threshold, the AI says:

"Hmm, I'm not sure..."

Air drawings are shakier, so we may lower it to 0.5 or 0.6 after camera tests.

Threshold	Answers	Right when it answers
0.5	98.1%	95.3%
0.6	96.7%	96.1%
0.7 (chosen)	95.1%	96.9%
0.8	93.4%	97.6%
0.9	91.0%	98.3%

Drawing in the air is hard

Line broke into dashes

Why: "index finger up" misread on a tilted hand

Fix: pinch to draw, with hysteresis: start below 0.30, stop above 0.45

One bad frame cut it

Why: tracker errors for a single frame

Fix: debounce: 2 steady frames to start, 6 to stop

Line jumped fingers

Why: tracker briefly mixed up fingers

Fix: skip jumps over 120 px; pen = thumb–index midpoint

Hand vanished

Why: hand lost for a few frames

Fix: keep the line alive for up to 8 missing frames

Jagged lines

Why: the fingertip shakes

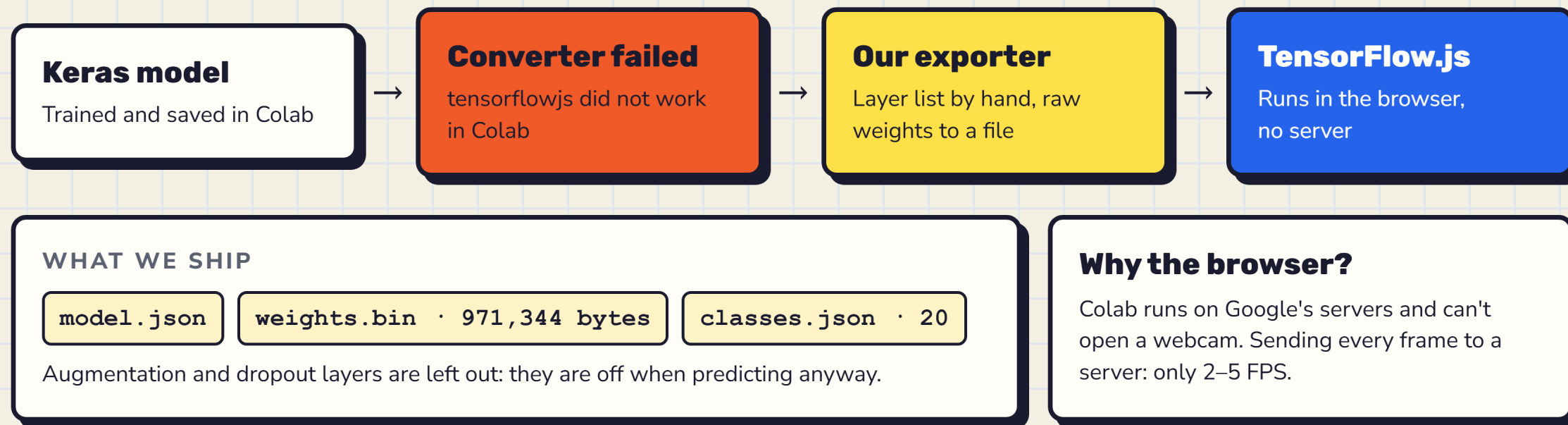
Fix: smoothing: new point = $0.6 \times \text{old} + 0.4 \times \text{new}$

Too thin at 28 × 28

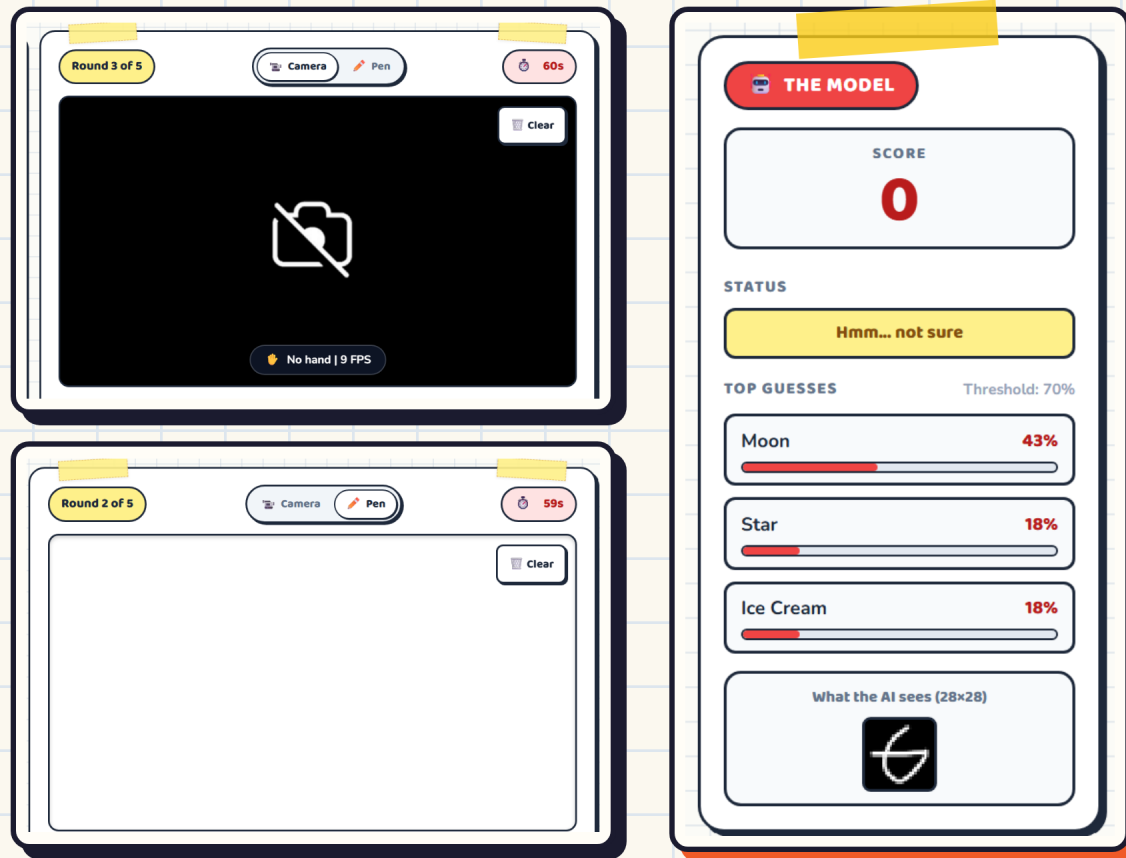
Why: a big drawing shrunk about 10×

Fix: 18 px line width, then crop, square, 10% margin

We wrote our own exporter



GuessWhat, ready to play



The AI's side, live

Top guess Moon at 43% is below the 70% threshold, so the AI says "Hmm... not sure". The 28 × 28 preview shows exactly what the model sees.

Live guesses + confidence bars

"What the AI sees"

Letter hints

5 rounds × 60 s

Leaderboard

Works on mobile

Built in Google Antigravity · hosted on GitHub + Vercel

Where GuessWhat could go next

01

AI literacy

Show students how a model sees, guesses and fails, in a fun way.

02

Events & museums

A touch-free booth anyone can play in a browser, no install.

03

Touch-free control

The same hand tracking can control screens and kiosks.

04

Better data

With consent, real air drawings could retrain the model.

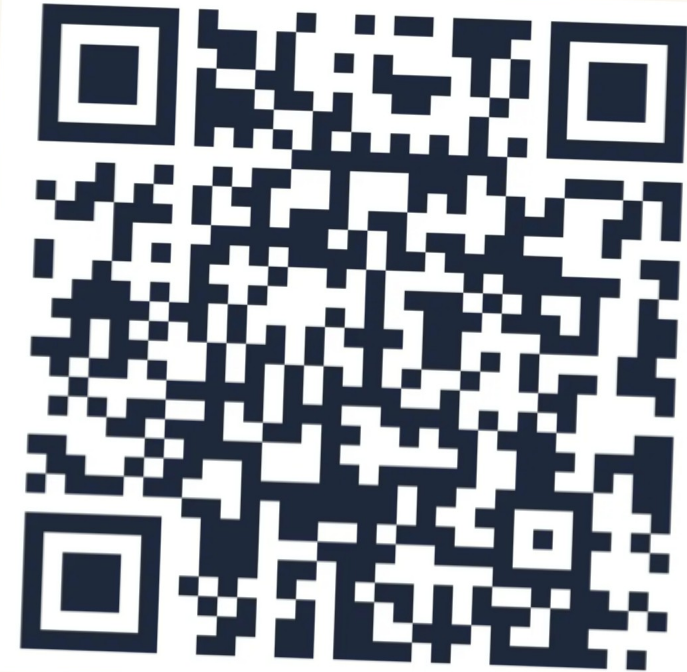
17 · LIVE DEMO

Your turn: beat the **AI**

PLAY guesswhat-five.vercel.app

CODE github.com/AstroRazan/GuessWhat

Thank you! · Razan Almasoud · Joud Alaskar · Hayat Fageeh



Scan to play